PROJECT SHOWCASE

I Built a Full-Stack AI PDF Chatbot

Turn any document into an interactive AI knowledge base using RAG

Next.js 16 | LangChain.js | Pinecone Vector DB | OpenAI | Tailwind CSS | Vercel

Zero Hallucinations

Answers grounded strictly in document chunks via vector similarity

Real-Time Streaming

Token-by-token response streaming with Vercel AI SDK.

Any Document Type

Tested on Resumes, Financial Invoices, Legal Contracts & Research Re

SYSTEM ARCHITECTURE

How the RAG Pipeline Works

From raw PDF binary to grounded, real-time answers in 4 stages:

01

PDF Ingestion & Text Extraction

PDFLoader parses the uploaded binary file into clean text pages with metadata.

02

Semantic Chunking

RecursiveCharacterTextSplitter divides text into 1,000-char chunks with 200-char overlap.

03

Vector Embeddings in Pinecone

OpenAI text-embedding-ada-002 indexes dense 1536-d vectors into a clean Pinecone index.

04

Context Retrieval & Streaming

Top-5 cosine similarity search fetches exact context; gpt-3.5-turbo streams the response.

USE CASE & DOCUMENT MAPPING

Screening Resumes & Engineering Candidates

Mapped Document: resume_priya_sharma.pdf

Chat Interface | Active PDF: resume_priya_sharma.pdf

User: "What are Priya Sharma's top AI/ML skills and accomplishments?"

AI Assistant (Grounded RAG Response):

- Core Skills: PyTorch, HuggingFace Transformers, LangChain, Pinecone, vLLM.
- Enterprise RAG: Designed knowledge base querying 500,000+ technical manuals.
- Accuracy & Quality: Reduced hallucination rates by 34% using Cohere Rerank.
- Fine-Tuning: Fine-tuned Llama-3 8B on legal data, achieving 91.4% accuracy.
- Education: M.Tech in AI from Indian Institute of Science (IISc), Bangalore.

INDUSTRY IMPACT

Recruiters spend an average of 7 minutes reviewing a technical CV.

With this RAG Chatbot, hiring managers can verify stack fit, production metrics, and degrees in under 5 seconds without manual scanning.

USE CASE & DOCUMENT MAPPING

Auditing Invoices & Financial Line Items

Mapped Document: invoice_apex_digital_services.pdf

Chat Interface | Active PDF: invoice_apex_digital_services.pdf

User: "Extract the invoice number, total amount due, and wire payment details."

AI Assistant (Grounded RAG Response):

- Invoice Number: INV-2026-4401 | Status: UNPAID
- Due Date: November 15, 2026 (Net 30 terms)
- Itemized Services: NVIDIA H100 Cluster Setup, RAG Knowledge Base, 24/7 SLA.
- Subtotal: $27,000.00 | Applicable Sales Tax (8.25%): $2,227.50
- Total Due: $29,227.50 | Bank: Silicon Tech Bank | Routing: STB1100234X

INDUSTRY IMPACT

Accounts Payable teams face expensive errors during manual line-item entry.

This app allows instant verification of tax rates, bank swift codes, and payment deadlines with zero hallucination.

USE CASE & DOCUMENT MAPPING

Reviewing Legal Contracts & Enterprise SLAs

Mapped Document: contract_saas_service_agreement.pdf

Chat Interface | Active PDF: contract_saas_service_agreement.pdf

User: "What uptime is guaranteed in the SLA and what are the downtime penalties?"

AI Assistant (Grounded RAG Response):

- Guaranteed Uptime: 99.95% monthly uptime excluding scheduled maintenance.
- SLA Penalty Level 1: If uptime drops below 99.5%, customer receives 15% credit.
- SLA Penalty Level 2: If uptime drops below 99.0%, customer receives 30% credit.
- Data Security: Customer Data remains exclusive property; SOC 2 & AES-256.
- Termination Notice: 60 days prior written notice required for cancellation.

INDUSTRY IMPACT

Legal and procurement teams can instantly spot compliance risks, penalty tiers, and governing jurisdictions without reading 40 pages of boilerplate clauses.

USE CASE & DOCUMENT MAPPING

Synthesizing Industry & Research Reports

Mapped Document: report_global_renewable_energy_2026.pdf

Chat Interface | Active PDF: report_global_renewable_energy_2026.pdf

User: "What were the top quantitative takeaways in renewable energy additions?"

AI Assistant (Grounded RAG Response):

- Capacity Record: Added 510 Gigawatts (GW) globally (+48% vs 2024 levels).
- Solar Leadership: Solar PV accounted for 72% of all new additions worldwide.
- Levelized Cost (LCOE): Utility-scale solar dropped to $0.028 per kWh.
- Storage: Battery Energy Storage Systems (BESS) surpassed 180 GWh.
- Regional Leader: APAC led with 58% of new solar installations.

INDUSTRY IMPACT

Executives and analysts no longer need hours to skim lengthy whitepapers.

Direct semantic retrieval pulls quantitative stats, benchmark percentages, and policy insights immediately.

OPEN SOURCE & LIVE DEMO

Ready to Explore the Code?

The complete repository is open-sourced on GitHub with full documentation.

GitHub Repository:

https://github.com/jayshreeganesh/pdf-chatbot

Includes 6 sample documents (Resumes, Invoices, Contracts, Reports) + USER_MANUAL.md

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